

Код для первого модуля

```
#include <linux/module.h>
#include <linux/kernel.h>

int init_module(void)
{
    pr_info("Test module loaded :D\n");
    return 0;
}

void cleanup_module(void)
{
    pr_info("Test module unloaded :(\n");
}

MODULE_LICENSE("GPL")
```

```
obj-m += lab1.o
#ccflags-y += -g -DEBUG

all:
    make -C /lib/modules/6.17.0-35-generic/build M=$(PWD) modules
clean:
    make -C /lib/modules/6.17.0-35-generic/build M=$(PWD) clean
~
~
```

Получаем lab1.ko

```
root@kivvi-VirtualBox:~/modules_lab/lab1# make
make -C /lib/modules/6.17.0-35-generic/build M=/root/modules_lab/lab1 modules
make[1]: Entering directory '/usr/src/linux-headers-6.17.0-35-generic'
make[2]: Entering directory '/root/modules_lab/lab1'
warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
You are using:          gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
CC [M] lab1.o
MODPOST Module.symvers
WARNING: modpost: missing MODULE_DESCRIPTION() in lab1.o
CC [M] lab1.mod.o
CC [M] .module-common.o
LD [M] lab1.ko
BTF [M] lab1.ko
Skipping BTF generation for lab1.ko due to unavailability of vmlinux
make[2]: Leaving directory '/root/modules_lab/lab1'
make[1]: Leaving directory '/usr/src/linux-headers-6.17.0-35-generic'
root@kivvi-VirtualBox:~/modules_lab/lab1# ls
lab1.c  lab1.mod  lab1.mod.o  Makefile      Module.symvers
lab1.ko lab1.mod.c lab1.o      modules.order
```

Загружаем и выгружаем модуль:

```
root@kivvi-VirtualBox:~/modules_lab/lab1# insmod lab1.ko
root@kivvi-VirtualBox:~/modules_lab/lab1# rmmod lab1.ko
```

И смотрим логи ядра:

```
root@kivvi-VirtualBox:~/modules_lab/lab1# dmesg | tail -n 5
[ 193.375040] 14:27:51.094704 Timer VBoxDRMClient: push screen layout data of 1 display(s)
to DRM stack, fPartialLayout=false, rc=VINIF_SUC
[ 1531.516518] lab1: loading out-of-tree module taints kernel.
[ 1531.516525] lab1: module verification failed: signature and/or required key missing - taint
ing kernel
[ 1531.516837] Test module loaded :D
[ 1543.279525] Test module unloaded :(
root@kivvi-VirtualBox:~/modules_lab/lab1#
```

Код для второго модуля:

```
#include <linux/module.h>
#include <linux/kernel.h>
#include <linux/fs.h>
#include <linux/rwlock.h>
#include <linux/uaccess.h>

static int major = 0;
static rwlock_t lock;
static char test_string[15] = "Hello!\n\0";

ssize_t test_read(struct file *fd, char __user *buff, size_t size, loff_t *off)
{
    size_t rc;
    read_lock(&lock);
    rc = simple_read_from_buffer(buff, size, off, test_string, 15);
    read_unlock(&lock);

    return rc;
}

ssize_t test_write(struct file *fd, const char __user *buff, size_t size, loff_t *off)
{
    size_t rc = 0;
    if (size > 15)
        return -EINVAL;

    write_lock(&lock);
    rc = simple_write_to_buffer(test_string, 15, off, buff, size);
    write_unlock(&lock);

    return rc;
}

static struct file_operations fops = {
    .owner = THIS_MODULE,
    .read = test_read,
    .write = test_write
};

int init_module(void)
{
    pr_info("Test module is loaded.\n");
    rwlock_init(&lock);
    major = register_chrdev(major, "lab2", &fops);

    if (major < 0)
        return major;
    pr_info("Major number is %d.\n", major);

    return 0;
}

void cleanup_module(void)
{
    unregister_chrdev(major, "lab2");
}

MODULE_LICENSE("GPL");
```

Собираем:

```
root@kivvi-VirtualBox:~/modules_lab/lab2# make
make -C /lib/modules/6.17.0-35-generic/build M=/root/modules_lab/lab2 modules
make[1]: Entering directory '/usr/src/linux-headers-6.17.0-35-generic'
make[2]: Entering directory '/root/modules_lab/lab2'
warning: the compiler differs from the one used to build the kernel
The kernel was built by: x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
You are using:          gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
CC [M]  lab2.o
lab2.c:11:9: warning: no previous prototype for 'test_read' [-Wmissing-prototypes]
  11 | ssize_t test_read(struct file *fd, char __user *buff, size_t size, loff_t *off)
      |         ^~~~~~
lab2.c:21:9: warning: no previous prototype for 'test_write' [-Wmissing-prototypes]
  21 | ssize_t test_write(struct file *fd, const char __user *buff, size_t size, loff_t *off)
      |         ^~~~~~
MODPOST Module.symvers
WARNING: modpost: missing MODULE_DESCRIPTION() in lab2.o
CC [M]  lab2.mod.o
CC [M]  .module-common.o
LD [M]  lab2.ko
BTF [M] lab2.ko
Skipping BTF generation for lab2.ko due to unavailability of vmlinux
make[2]: Leaving directory '/root/modules_lab/lab2'
make[1]: Leaving directory '/usr/src/linux-headers-6.17.0-35-generic'
root@kivvi-VirtualBox:~/modules_lab/lab2# ls
lab2.c  lab2.mod  lab2.mod.o  Makefile  Module.symvers
lab2.ko lab2.mod.c lab2.o      modules.order
```

Загружаем:

```
root@kivvi-VirtualBox:~/modules_lab/lab2# insmod lab2.ko
root@kivvi-VirtualBox:~/modules_lab/lab2# dmesg | tail -n 5
[ 4063.508504] 15:32:21.844960 Timer      VBoxDRMClient: push screen layout data of 1 display(s)
to DRM stack, fPartialLayout=false, rc=VINF_SUC
[ 4063.758293] 15:32:22.094582 Timer      VBoxDRMClient: push screen layout data of 1 display(s)
to DRM stack, fPartialLayout=false, rc=VINF_SUC
[ 4064.007988] 15:32:22.344470 Timer      VBoxDRMClient: push screen layout data of 1 display(s)
to DRM stack, fPartialLayout=false, rc=VINF_SUC
[ 4904.453022] Test module is loaded.
[ 4904.453029] Major number is 239.
root@kivvi-VirtualBox:~/modules_lab/lab2#
```

```
root@kivvi-VirtualBox:/dev# mknod lab2 c 239 0
root@kivvi-VirtualBox:/dev# ls -la lab2
crw-r--r-- 1 root root 239, 0 Jun 13 11:53 lab2
root@kivvi-VirtualBox:/dev#
```

```
root@kivvi-VirtualBox:/dev# cat lab2
cat: lab2: Bad address
root@kivvi-VirtualBox:/dev# echo -n "test" > /dev/lab2
root@kivvi-VirtualBox:/dev# cat lab2
cat: lab2: Bad address
root@kivvi-VirtualBox:/dev#
```

Эту проблему с Bad address я на Linux Mint с ядром 6.17.0-35 так и не смог решить, пробуя переписывать код, поэтому перешел на Ubuntu

make

```
root@kivvi-VirtualBox:~/modules_lab/lab2# make
make -C /lib/modules/5.15.0-139-generic/build M=/root/modules_lab/lab2 modules
make[1]: Entering directory '/usr/src/linux-headers-5.15.0-139-generic'
  CC [M] /root/modules_lab/lab2/lab2.o
  MODPOST /root/modules_lab/lab2/Module.symvers
  CC [M] /root/modules_lab/lab2/lab2.mod.o
  LD [M] /root/modules_lab/lab2/lab2.ko
  BTF [M] /root/modules_lab/lab2/lab2.ko
Skipping BTF generation for /root/modules_lab/lab2/lab2.ko due to unavailability of vmlinux
make[1]: Leaving directory '/usr/src/linux-headers-5.15.0-139-generic'
root@kivvi-VirtualBox:~/modules_lab/lab2#
```

Загрузка

```
make[1]: Leaving directory '/usr/src/linux-headers-5.15.0-139-generic'
root@kivvi-VirtualBox:~/modules_lab/lab2# ls
lab2.c  lab2.mod  lab2.mod.o  Makefile      Module.symvers
lab2.ko lab2.mod.c lab2.o      modules.order
root@kivvi-VirtualBox:~/modules_lab/lab2# insmod lab2.ko
root@kivvi-VirtualBox:~/modules_lab/lab2# dmesg | tail -n 5
[ 1426.606187] 06:13:36.025184 Timer      VBoxDRMClient: push screen layout data of 1
display(s) to DRM stack, fPartialLayout=false, rc=VINIF_SUC
[ 2655.528301] lab2: loading out-of-tree module taints kernel.
[ 2655.528334] lab2: module verification failed: signature and/or required key missi
ng - tainting kernel
[ 2655.528711] Test module is loaded.
[ 2655.528714] Major number is 237.
root@kivvi-VirtualBox:~/modules_lab/lab2#
```

```
root@kivvi-VirtualBox:/dev# mknod lab2 c 237 0
root@kivvi-VirtualBox:/dev# ls -la lab2
crw-r--r-- 1 root root 237, 0 jun 14 08:36 lab2
root@kivvi-VirtualBox:/dev# cat lab2
Hello!
root@kivvi-VirtualBox:/dev#
```